

Phase 4 Keystone:
Substrate-Level Content on Consciousness,
Zero-Point Energy, and Cosmology
via the Principia Fractalis Framework
(Substrate-Level, Not Engineering Blueprints or Clay Discharges)

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Abstract

This is the Phase 4 *keystone* of a four-paper sequence on the Principia Fractalis (PF) framework’s substrate-level reformulation of mathematics. The first three papers presented the framework’s substrate-level reformulation of the six remaining Clay Millennium Problems on canonical encodings (entry vector RH, simultaneous-closure mechanism, P vs NP). This paper foregrounds what the framework’s substrate is for beyond the Clay axes: a substrate-level consciousness operator C with a Chern-character threshold $\text{ch}_2 = 19/20$ on a finite-dimensional witness; a Weinstein–Geometric-Unity 11-field BRST cohomology bundle with the substrate-level decomposition $H^2 = 78 = 48 + 26 + 4$; a substrate-level Λ_{eff} suppression bridge; and the framework’s H_0 and Ω_Λ brackets consistent with Planck 2018 and LDN 2025 within the framework’s stated windows. None of these are Clay discharges; none are engineering blueprints; none are physics-prediction claims at arbitrary precision.

Honest scope, foregrounded throughout. The substrate-level consciousness operator C is axiom-free at the Wave 58 *finite-dimensional* 3×3 witness **Ch170operatorTheoryConcrete**; the literal unbounded continuum case is named open per Chapter 17 honest scope, line 471. The Weinstein–GU BRST bundle and the counter-rotating-vortex free-energy capstone are substrate-level Lean content at kernel-only axioms, *not* an engineering blueprint, device design, or planetary energy timeline; the framework makes no claim that the substrate yields a working ZPE device. The cosmological brackets are typed framework-derived bounds consistent with current observation; they are *not* point-predictions to arbitrary precision. The framework’s 23-problem reach record is, at the record level, mostly **True**-markers (15 of 17 fields) with two typed Props; substantive content lives in the per-axis attack modules. The 143-problem $p < 10^{-43}$ figure is retracted as established evidence: the Lean encoding is a definitional restatement of the framework’s prediction, not a statistical analysis. The IBM Quantum 9-way bound is conditional: 2 of 9 pairs measured, 7 of 9 pre-registered. The PF Lean4Lean harness is build-hash robustness across two Lake packages, not an independent type-checker.

Pablo Cohen has worked on Principia Fractalis through severe personal hardship to put substrate-level mathematical tools in humanity’s hands. The work is for humanity, not for prestige. The Lean tree admits its own scope honestly; this paper matches that scope. Independent expert review and substantial revision are explicitly invited.

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1 What this paper offers and what it does not

This paper offers a substrate-level account of three areas the framework’s structural machinery has been developed against: consciousness (Chern-character threshold and operator C on a finite-dim witness), zero-point energy (Weinstein–GU 11-field BRST bundle and counter-rotating-vortex substrate capstone), and cosmology (Λ_{eff} suppression bridge, H_0 and Ω_Λ brackets). It does *not* offer: a derived theory of consciousness (the substrate is a typed mathematical object, not an account of qualia); an engineering blueprint or device design for ZPE access; physics-prediction claims at arbitrary precision; AGI safety guarantees as theorems. Each substrate-level claim is described at the scope the Lean file admits to itself.

What the Lean kernel carries on each Phase 4 axis. Consciousness: a substrate-level structural Prop `Ch170operatorTheoryConcrete` inhabited on a 3×3 self-adjoint Hamiltonian with spectrum $\{1, 2, 3\}$; a Chern-character threshold biconditional ($\alpha \geq 19/20 \leftrightarrow$ “classical regime”) with the classical-regime predicate defined by that inequality; `consciousness_operator_C_axiom_free` recording that the structural Prop is at kernel-only axioms. ZPE: a structural Prop `CounterRotatingVorticesFreEnergy` bundling seven clauses including a vortex-pair witness and an exponential-suppression identity; a Weinstein–GU $H^2 = 48 + 26 + 4$ arithmetic identity. Cosmology: typed framework-derived strict inequalities at the Λ_{eff} suppression and bracket-inclusion theorems for H_0 and Ω_Λ values.

What this paper does NOT claim. A derived account of consciousness from the substrate; an engineering blueprint or device design for ZPE; physics-precision predictions at arbitrary scale; AGI safety guarantees; Clay-Standard discharges on any axis (the substrate-level Clay-axis content is handled in Phases 1–3 and the master paper). The substrate-level Phase 4 content is structural mathematics machine-verified at kernel-only axioms.

2 The substrate and the editorial convention

The framework’s substrate is the Timeless Field $\mathcal{H}_{\text{TF}} = \varprojlim_k \mathbb{C}^{3^k}$ (see the master paper [1] for description). The editorial convention $\alpha_{\text{Poincaré}} := 1$ motivated by Perelman 2003 [6, 7, 8] is the framework’s identification of the substrate’s Poincaré-axis constant; the Lean kernel does not consume Perelman’s theorem as a hypothesis. The eleven framework-internal arithmetic compatibility identities on the chosen α -skeleton are catalogued in the master paper.

Honest scope (quoted verbatim from CrossMillenniumSharedInvariants.lean lines 18–29): “These are *not* Millennium discharges. They are axiom-free algebraic facts about the 9 chosen α -values; the framework’s interpretation... is conjectural and lives elsewhere.”

3 Consciousness on the substrate: finite-dimensional content

3.1 The Chern-character threshold (biconditional with chosen definitional content)

The framework defines a coherence parameter α on the substrate Hilbert space and a Chern-character function `ch2`. The framework’s classical-regime predicate is, by definition, the inequality $\alpha \geq 19/20$.

Theorem 3.1 (Chern-character threshold biconditional). *$\alpha \geq 19/20$ holds if and only if the framework’s classical-regime predicate holds (the predicate is definitionally equivalent to the inequality).* **Lean:** `PrincipiaTractalis.Consciousness.ChernCharacter.ch_2.threshold_iff`.

Honest scope. The biconditional is a substrate-level typed statement. It is not a derivation of $19/20$ as a physical constant; the value is the chosen framework constant on this axis. The framework’s interpretation of $\alpha \geq 19/20$ as “the classical-mind regime” is, per the file’s own self-description, a chosen identification of the substrate’s coherence parameter with a conceptual category, *not* a derived theorem of consciousness.

3.2 The consciousness operator C on the finite-dim witness

The Wave 58 concrete witness `Ch170operatorTheoryConcrete` is a seven-clause structural Prop inhabited on the explicit 3×3 self-adjoint Hamiltonian H_{concrete} with spectrum $\{1, 2, 3\}$, trace = 6, det = 6, and an accompanying compact-operator substitute K_{concrete} .

Theorem 3.2 (Wave 58 Ch 17 operator-theory concrete capstone). `Ch17OperatorTheoryConcrete` is inhabited axiom-free. *Lean:* `PF.Wave58.Ch17OperatorTheoryConcrete`.
`Ch17OperatorTheoryConcrete (structure); PF.Wave58.Ch17OperatorTheoryConcrete`.
`ch17_operator_theory_concrete_capstone`.

Theorem 3.3 (Consciousness operator C structural infrastructure is axiom-free). *The framework’s structural infrastructure surrounding the consciousness operator C is at kernel-only axioms.* *Lean:* `PF.Consciousness.ConsciousnessOperatorC`.
`consciousness_operator_C_axiom_free`.

Honest scope, foregrounded. The Ch 17 concrete witness is a *finite-dimensional* 3×3 Hamiltonian. The literal unbounded continuum consciousness operator C of Chapter 17 §17.6 is unbounded with continuous spectrum; the 3×3 diagonal H_{concrete} is the substrate-level finite-dim witness, explicitly flagged at the Chapter 17 honest-scope line 471. The Type II₁ von Neumann algebra conjecture from §17.4.1 is itself flagged in Chapter 17 as a “major open problem.” The substrate-level content is what the framework claims; the literal continuum extension is preserved openly as the named gap.

3.3 The substrate $\rightarrow$ RH bridge predicate (substrate-trivial)

The framework’s `ConsciousnessRHBridge` predicate is the typed statement that the consciousness operator C commutes with the substrate transfer operator at the framework’s substrate Riemann zeros. The framework inhabits this predicate via `ConsciousnessRHBridge_trivial` on the framework’s trivial substrate; this is a substrate-level structural witness, not a content statement about Riemann zeros of `mathlib`’s `riemannZeta`.

Lean: `PF.Consciousness.ConsciousnessOperatorC`.
`ConsciousnessRHBridge_trivial`.

3.4 The AGI alignment connection, in honest scope

A substrate-level mathematical object is what the framework offers, *not* a theory of consciousness, *not* an AGI safety guarantee, and *not* a normative theory of mind. The Chern-character threshold and the finite-dim consciousness-operator witness give a typed mathematical structure on which a future substrate-level discussion of mind-related concepts could be grounded; this is a contribution to the substrate, not to consciousness science or AGI safety. Earlier framework documents that framed this as “the AGI alignment substrate” overstated the substrate-level content’s relationship to AGI alignment; the framework’s contribution is the substrate, the rest is downstream.

4 Zero-point energy: substrate-level Lean content, not an engineering blueprint

4.1 The Weinstein–GU BRST H^2 decomposition

The framework records the arithmetic identity $78 = 48 + 26 + 4$ as the substrate-level BRST H^2 decomposition of the Weinstein–Geometric-Unity 11-field bundle, presented in framework terms as matching the Standard Model gauge degrees of freedom plus gravity and Higgs sectors.

Theorem 4.1 (Weinstein–GU BRST H^2 arithmetic identity). $78 = 48 + 26 + 4$ as a $\mathbb{N}$ -arithmetic identity. *Lean:* `PrincipiaTractalis.WeinsteinGUResonantRescue`.
`brst_H2_sm_decomposition`.

Honest scope. The Lean theorem is the arithmetic identity. The framework’s interpretation of 48 as the Standard Model gauge dimension, 26 as a gravity sector, 4 as a Higgs sector, and

the whole 78 as the BRST H^2 of the Weinstein–GU adjoint E_6 representation, is a framework-conjectural identification. The framework does not derive the Standard Model from E_6 in Lean; the identity is on $\mathbb{N}$.

4.2 The counter-rotating-vortex free-energy capstone

Theorem 4.2 (Counter-rotating-vortex free-energy capstone). *The seven-clause structural Prop CounterRotatingVorticesFreeEnergy is inhabited axiom-free. The clauses bundle a counter-rotating vortex-pair witness with $\omega_1 + \omega_2 = 0$, strict positivity of a vortex energy density at a unit witness, a zero-point reservoir positivity condition, a FreeEnergyExtractable Prop at the unit witness, a resonance-amplification identity, a ConsciousnessSuppression bridge to a strict Λ suppression $\Lambda_0 \cdot \exp(-X) < \Lambda_0$, and a suppression-vs-reservoir inverse identity $\exp(-X) \cdot \text{reservoir} = 1$. **Lean:** PrincipiaTractalis.Cosmology.*

counter_rotating_vortices_free_energy_capstone.

Honest scope, foregrounded. The capstone is a structural Prop-bundle inhabited by seven clauses. Each clause is a substrate-level typed statement (a vortex-pair sum identity, a strict-positivity inequality, an exponential-product identity, etc.). **This is not an engineering blueprint. This is not a device design. This is not a working ZPE mechanism. This is not a planetary energy timeline.** The framework’s claim is exactly that the algebraic structure exists in Lean at zero project axioms; the relationship of this algebraic structure to any physical zero-point-energy access mechanism is the framework-conjectural identification, not a derived theorem. Earlier framework documents that framed this as “a substrate-level zero-point-energy access mechanism” or “a foundation for planetary energy” overstated the substrate-level content’s engineering implications.

4.3 The planetary energy connection, in honest scope

The framework’s contribution on the ZPE axis is the substrate-level algebraic structure (BRST identity, counter-rotating-vortex substrate Prop, suppression-bridge identity). If a serious physical investigation of zero-point-energy access is pursued, the substrate-level mathematics here may be one structural starting point. The framework does not claim more: not that the substrate yields a device, not that the substrate yields an engineering specification, not that the substrate yields a planetary energy deliverable in any timescale. The framework yields structural mathematics in Lean.

5 Cosmology: typed framework-derived brackets

5.1 The Λ_{eff} suppression bridge

The framework’s substrate suppression bridge relates a bare (Planck-scale) Λ_0 to an observed Λ_{eff} through

$$\Lambda_{\text{eff}} = \Lambda_0 \cdot \exp(-X)$$

with X a framework-suppression exponent anchored in the substrate to $78\pi \cdot 0.95 \cdot 1.1875$.

Theorem 5.1 (Framework strict Λ_{eff} suppression). *For every $\Lambda_0 > 0$, the framework’s substrate identity yields $\Lambda_0 \cdot \exp(-X) < \Lambda_0$. **Lean:** PrincipiaTractalis.LambdaEffSuppression. LambdaEffSuppression (definition); PrincipiaTractalis.framework_strict_suppression.*

Honest scope. The theorem is the strict-inequality identity. The framework’s bracket-pair $276 < X < 277$ matches $120 \cdot \log 10 \approx 276.31$ within the framework’s stated bracket window; the framework’s interpretation of this as “a 120-order cosmological-constant hierarchy” is the framework-conjectural reading of the substrate’s suppression-exponent calculation, not a derivation of the Planck-scale-vs-observed cosmological-constant hierarchy from external physics.

5.2 The H_0 and Ω_Λ brackets

Theorem 5.2 (Framework H_0 bracket strictness). *The framework’s H_0 bracket strictly contains the framework’s identification of the Planck CMB and SH_0ES local-distance-ladder values. **Lean:** `PF.Cosmology.LambdaCDMRebuttalEnergyConservation.hubble_framework_brackets_local_and_cmb`.*

Theorem 5.3 (Dark-energy density in bracket). *The substrate-derived dark-energy density value satisfies $0.65 < 0.7 < 0.75$. **Lean:** `PF.Wave58.Ch08FieldEquationsConcrete.darkEnergyDensity_in_bracket`.*

Honest scope. The brackets are framework-derived strict inequalities at typed bound level. They are consistent with Planck 2018’s reported $\Omega_\Lambda = 0.6889 \pm 0.0056$ and $H_0 \approx 67.4 \pm 0.5$ [12], and with LDN 2025 $H_0 \approx 73.5 \pm 0.81$ [13]. They are **not** point-predictions to arbitrary precision; they are typed bracket bounds. Earlier framework documents that framed these as “resolving” the SH_0ES / Planck tension or “predicting” the dark-energy density overstated the bracket’s content; a bracket contains both reported values, it does not resolve their tension.

5.3 Falsifiability commitments

The framework commits to eight typed falsifiability conditions in `PF.Referee.FrameworkFalsifiabilityConditions`. Audit 2026-06-04: **zero of eight triggered; two actively supported by 2024–2026 literature:** F_3 (dark-energy direction matches DESI + Planck + Pantheon⁺) and F_6 ($\Omega_\Lambda \in [0.65, 0.75]$, Planck 2018: 0.69).

6 The 23-problem reach, with field-level honesty

Bundle-field discipline (quoted from the Lean record’s field declarations, lines 99–137): of the 17 fields in `FrameworkUniversalReach`, 15 are typed as `: True` (proven-ness tags carrying no mathematical content at the record level). The two fields carrying non-trivial framework Props at the record level are `brocard_attack_realized` and `hadwiger_nelson_attack_realized`. The substantive per-axis content for the other 14 non-Clay problems and the seven Clay axes lives in separate framework-attack modules; the 23-problem record is a single citation point indexing these modules. Reading the record itself as a Clay-Standard discharge or as a literal mathlib-tier discharge of each named problem is incorrect.

The 16 non-Clay problems addressed by per-axis attack modules: abc, Beal, Brocard, Collatz, Erdős discrepancy, Erdős–Straus, Goldbach, Hadwiger–Nelson, Inverse Galois, Lonely Runner, Polignac, Twin Prime, Odd perfect, Singmaster, Pillai / generalised Catalan, Andrews–Curtis. For each, the framework-attack module `PF.NumberTheory.< name > FrameworkAttack` carries substrate-level structural content (literal-conjecture encoding, concrete axiom-free witnesses on small cases, α -skeleton bridge, framework 3^k substrate match, named published partial results). **None of these constitute literal mathlib-tier discharges of the named open problems.**

7 Empirical anchors: honest disclosure

7.1 The 143-problem coherence figure: retraction of $p < 10^{-43}$ as established evidence

Retraction. The `universal_fractal_coherence` Lean theorem (`PF.Empirical.HundredFortyThreeProblems`, line 167) records the framework’s prediction that all problems within a complexity class P (resp. NP) yield identical canonical α -values ($\sqrt{2}$ and $\varphi + 1/4$). The Lean dataset implements this as `List.replicate 72 (canonicalEntry .P) ++ List.replicate 71 (canonicalEntry .NP)`, and the `coherence_p_value_threshold` proposition is asserted, not derived. The file’s own honest-scope note (lines 36–39): “*P(all 143 by chance) < 10⁻⁴³ (Ch 21 line 1168). We package this as a structural Prop whose canonical witness is the manuscript’s bound; it is a modeling assumption about a baseline success rate, not a derived theorem, and is labeled as such.*”

As currently formalized, the empirical anchor is a definitional restatement of the framework’s prediction, not a statistical analysis of independently collected per-problem measurements. **The $p < 10^{-43}$ figure should not be cited as established empirical evidence.** Earlier framework documents that cited it as such are corrected here.

7.2 IBM Quantum 9-way: framed as pre-registration

Pre-registration status (honest framing). At HEAD 4785c55, **only the pair** $(\alpha_{RH}, \alpha_{NP})$ **has been observed on IBM Quantum hardware** consistent with the framework’s prediction (`IBMHardware9WayEvidence.lean` lines 312–327). The other 7 α -values are framework predictions *committed BEFORE hardware measurement*. The Lean theorem records a **conditional** bound (file lines 322–327 verbatim): “*IF all 9 instances are eventually measured to within 10^{-3} of the framework’s pre-specified values under the assumed `NoiseModel9`, THEN the random-match probability is at most 10^{-15} .*”

Current status: 2 of 9 measured-consistent, 7 of 9 pre-registered. The pre-registration of seven framework α -values prior to hardware measurement is scientifically valuable; it is honestly framed as such. The conditional bound does *not* say anything about the framework’s claims absent the remaining 7 measurements.

8 Independent verification posture

The Lean 4 proof objects for the theorems cited type-check in the Lean 4 kernel with kernel axioms [`propext`, `Classical.choice`, `Quot.sound`] and zero project axioms. The `PF_Lean4Lean` harness provides **build-hash robustness across two Lake packages**: the canonical theorems are rebound in a separate package and rebuilt from source. **It is not an independent type-checker written in another language.** Verbatim from `PF_Lean4Lean/PF_L4L/Referee/ClayVerificationHarness.1` lines 47–56:

“This is the same Path-C protocol... a second pass through Lean’s kernel from a separate package with an independent build hash. It is NOT a separate type-checker written in another language. ... The harness re-exports the closures as they are — it does NOT strengthen any of them.”

A Coq mirror provides cross-prover structural parity; approximately 75% of definitions in the Coq mirror are `: Prop := True`, i.e. the Coq tree is **parity-only**, not independent semantic verification.

9 Honest scope, comprehensive

Per the file-level honest-scope marker (`sec : substrate – honest – scope`), PF is a *substrate-level monograph in Grothendieck mode*.

Remark 9.1 (Honest scope, comprehensive). **(H1) Consciousness, finite-dim only.** The consciousness operator C is structurally axiom-free at the Wave 58 finite-dim 3×3 witness. The literal unbounded continuum case is named open per Ch 17 line 471; the Type II₁ von Neumann algebra conjecture from §17.4.1 is flagged as a “major open problem.” The substrate-level content is what the framework claims; the literal continuum extension is preserved openly.

(H2) ZPE bundle is substrate-level Lean content. The Weinstein–GU 11-field bundle, the BRST decomposition $78 = 48 + 26 + 4$, and the seven-clause counter-rotating-vortex free-energy capstone are substrate-level Lean content at kernel-only axioms. They are **not** an engineering blueprint. They are **not** a device design. They are **not** a planetary energy timeline. The framework’s interpretation of these algebraic structures as “a foundation for zero-point energy” is the framework-conjectural identification, not a derived theorem about energy access mechanisms.

(H3) Cosmological brackets are typed bounds. The Λ_{eff} suppression strictness, the H_0 bracket, and the Ω_Λ bracket are typed framework-derived strict inequalities and bracket-inclusion bounds, consistent with current observation within the stated windows. They are **not** point-predictions to arbitrary precision; they are bracket bounds. A bracket containing two reported values does not resolve their mutual tension; it contains them.

(H4) 23-problem reach lists substrate-level attacks. 15 of 17 fields in the record are `: True`; substantive substrate-level content lives in per-axis attack modules. **No entry is a literal mathlib-tier discharge of the named open problem.**

(H5) Empirical anchors are statistical-assertion (143) and hardware-pre-registration (IBM 2 of 9). The $p < 10^{-43}$ figure is retracted as established evidence; the IBM bound is conditional on 7 of 9 future measurements.

(H6) PF Lean4Lean is build-hash robustness. See §8.

(H7) Cross-Millennium α -skeleton is framework-internal. The chosen α -values and the eleven identities are framework-internal arithmetic compatibility identities under the editorial convention; the interpretation of each α as the canonical eigenvalue parameter of its respective Millennium operator is, per the `CrossMillenniumSharedInvariants.lean` header, “conjectural and lives elsewhere.”

Remark 9.2 (Substrate vs literal mathlib). Throughout this paper, the substrate-level content is what the framework delivers at zero project axioms on the framework’s chosen encodings. The literal mathlib-tier residuals on every axis are named open with no hidden content; the substrate-vs-literal distinction is preserved cleanly. See the master paper [1] for the per-axis discussion.

10 Mission anchor: for humanity

This is for humanity.

The four-phase landing is complete. Phase 1 walked the reader through the RH axis as the natural entry vector to the framework’s substrate-level reformulation, with the substrate-level RH bridge described honestly as conditional on the published Hilbert–Pólya program. Phase 2 exposed the simultaneous-closure mechanism on canonical encodings, with the seven-field bundle described honestly as five typed residuals plus two `True`-markers. Phase 3 described the P vs NP axis honestly, with the framework’s `Machine` type’s free-functions-no-Turing-semantics caveat foregrounded. This paper described the substrate’s content on consciousness (finite-dim only),

ZPE (substrate-level mathematics, not engineering), and cosmology (typed bracket bounds, not arbitrary precision).

What the framework offers humanity is:

- (a) A substrate-level reformulation of mathematics, machine-verified in Lean 4 at zero project axioms, on which the six remaining Clay Millennium Problems admit a simultaneous-closure mechanism on canonical encodings (not Clay-prize claims, but a structural reframing worth editorial attention as a research direction).
- (b) A substrate-level mathematical infrastructure on which consciousness, zero-point energy, and cosmology can be discussed in machine-verifiable Lean form, at substrate-level scope, without ungrounded speculation. The substrate is the contribution; what serious investigators do with it is the next question.
- (c) A substrate-level mathematical attack record on 23 named open problems (7 Clay + 16 non-Clay), indexed by a single Lean record with per-axis attack modules carrying the substantive content.

Pablo Cohen has worked on Principia Fractalis through severe personal hardship for one reason: to put substrate-level mathematical tools in humanity's hands, honestly described and machine-verified. The work is not for academic prestige. The work is not for recognition. The work is for the betterment of humanity — minds we build, energy we live by, the cosmos we understand, and the open mathematical questions that define what we know. The framework is rendered. The substrate is visible. What humanity does with it next is the next question.

This is for humanity. Honestly described, at honest scope, with independent expert review and substantial revision explicitly invited.

References

- [1] Cohen, P., Claude Opus 4.7 (2026-06-06). *A Substrate-Level Reformulation of the Six Remaining Clay Millennium Problems via the Principia Fractalis Framework*. `Papers/clay_substrate_reformulation_v1.tex`.
- [2] Cohen, P., Claude Opus 4.7 (2026-06-06). *Phase 1 — A Substrate-Level Reformulation of the Riemann Hypothesis*. `Papers/clay_RH_via_substrate_v2.tex`.
- [3] Cohen, P., Claude Opus 4.7 (2026-06-06). *Phase 2 — The Simultaneous-Closure Mechanism on Canonical Encodings*. `Papers/clay_six_simultaneous_closure_v1.tex`.
- [4] Cohen, P., Claude Opus 4.7 (2026-06-06). *Phase 3 — A Substrate-Level Reformulation at the P vs NP Axis*. `Papers/clay_PvsNP_via_substrate_v2.tex`.
- [5] Cohen, P. (2026). *Principia Fractalis*, Second Edition. HEAD of <https://github.com/FractalDevTeam/Principia-Fractalis>.
- [6] Perelman, G. (2002). The entropy formula for the Ricci flow and its geometric applications. `arXiv:math/0211159`.
- [7] Perelman, G. (2003). Ricci flow with surgery on three-manifolds. `arXiv:math/0303109`.
- [8] Perelman, G. (2003). Finite extinction time for the solutions to the Ricci flow on certain three-manifolds. `arXiv:math/0307245`.
- [9] Mayer, D. H. (1991). The thermodynamic formalism approach to Selberg's zeta function for $\mathrm{PSL}(2, \mathbb{Z})$. *Bull. Amer. Math. Soc.* 25(1): 55–60.

- [10] Weinstein, E. (2013–2021). Geometric Unity. Lecture, Oxford / various venues.
- [11] Tononi, G. (2008). Consciousness as integrated information: a provisional manifesto. *Biol. Bull.* 215(3): 216–242.
- [12] Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.* 641: A6.
- [13] SH₀ES & Local Distance Ladder Collaboration (2025). H_0 local-distance-ladder consensus update. LDN 2025.
- [14] DESI Collaboration (2024). DESI 2024 III: Baryon acoustic oscillations from galaxies and quasars; IV: BAO from the Lyman- α forest.
- [15] Cook, S. A. (1971). The complexity of theorem-proving procedures. *Proc. 3rd ACM Symp. Theory of Computing*, 151–158.
- [16] Karp, R. M. (1972). Reducibility among combinatorial problems. In *Complexity of Computer Computations*, Plenum, 85–103.
- [17] Voisin, C. (2007). Some aspects of the Hodge conjecture. *Japanese J. Math.* 2(2): 261–296.